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import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

dataset=pd.read_csv(r"C:\Users\srini\Downloads\Investment.csv")

x=dataset.iloc[:, :-1]

y=dataset.iloc[:, 4]

x=pd.get_dummies(x, dtype=int)

from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=0)

from sklearn.linear_model import LinearRegression
regressor=LinearRegression()
regressor.fit(x_train,y_train)

y_pred=regressor.predict(x_test)

print(regressor.coef_)

print(regressor.intercept_)

x=np.append(arr=np.full((50,1),42467).astype(int),values=x,axis=1)

import statsmodels.api as sm
x_opt=x[:,[0,1,2,3,4,5]]

# Ordinary least squares

regressor_OLS=sm.OLS(endog=y,exog=x_opt).fit()
regressor_OLS.summary()

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